SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

I [redacted] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: [redacted]

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	2	1	1	1	1
Method Selection	20%	2	2	2	2	2
Solution Process	25%	2	2	2	2	2
Accuracy of Calculation	20%	1	1	1	1	1
Interpretation of Results	15%	1	1	1	1	1

Total Marks 36 /40

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

I. Introduction

Subnetting is the process of dividing a large network into smaller logical networks called subnets. It helps in efficient IP address allocation, reduces broadcast traffic, and improves network management.

The university is assigned the IP address block 192.168.20.0/24 and needs to allocate subnets to three laboratories using fixed subnet masks:

- Lab A: /26
- Lab B: /27
- Lab C: /28

II. Given Data

Particulars	Details
Original network address	192.168.20.0/24
Original subnet mask	255.255.255.0
Lab A prefix	/26
Lab B prefix	/27
Lab C prefix	/28

III. Subnetting Formulas

1. Number of host bits = 32 − Prefix length
2. Total IP addresses = 2^{host bits}
3. Usable host addresses = 2^{host bits}- 2
4. Block size = 256 − subnet mask value in the last octet

   Two addresses are excluded from usable hosts: the network address and the broadcast address.

IV. Calculation for Lab A – /26

Step 1: Find the subnet mask

A /26 prefix means 26 bits are used for the network and 6 bits for hosts.

Subnet mask: 255.255.255.192

Step 2: Calculate the block size

Block size =256-192=64

Step 3: Calculate the number of addresses

Total addresses=2^6=64

Usable hosts=64-2=62

Lab A is allocated the first 64 addresses from the given network.

- Network address: 192.168.20.0
- First usable host: 192.168.20.1
- Last usable host: 192.168.20.62
- Broadcast address: 192.168.20.63

V. Calculation for Lab B – /27

Step 1: Find the subnet mask

A /27 prefix means 27 bits are used for the network and 5 bits for hosts.

Subnet mask: 255.255.255.224

Step 2: Calculate the block size

Block size=256-224=32

Step 3: Calculate the number of addresses

Total addresses=2^5=32

Usable hosts=32-2=30

The next available subnet begins at 192.168.20.64.

- Network address: 192.168.20.64
- First usable host: 192.168.20.65
- Last usable host: 192.168.20.94
- Broadcast address: 192.168.20.95

VI. Calculation for Lab C – /28

Step 1: Find the subnet mask

A /28 prefix means 28 bits are used for the network and 4 bits for hosts.

Subnet mask: 255.255.255.240

Step 2: Calculate the block size

Block size=256-240=16

Step 3: Calculate the number of addresses

Total addresses=2^4=16

Usable hosts=16-2=14

The next available subnet begins at 192.168.20.96.

- Network address: 192.168.20.96
- First usable host: 192.168.20.97
- Last usable host: 192.168.20.110
- Broadcast address: 192.168.20.111

VII. Summary Table

The following table shows the subnet allocation, assuming the labs are assigned consecutive, non-overlapping subnets starting from 192.168.20.0.

Details	Lab A	Lab B	Lab C
CIDR prefix	/26	/27	/28
Subnet mask	255.255.255.192	255.255.255.224	255.255.255.240
Network address	192.168.20.0	192.168.20.64	192.168.20.96
First usable host	192.168.20.1	192.168.20.65	192.168.20.97
Last usable host	192.168.20.62	192.168.20.94	192.168.20.110
Broadcast address	192.168.20.63	192.168.20.95	192.168.20.111
Total IP addresses	64	32	16
Usable host addresses	62	30	14

VIII. Total Usable Host Addresses

The total number of usable host addresses across the three labs is:

62+30+14=106

Therefore, the three labs together have 106 usable host IP addresses.

IX. Conclusion

The network 192.168.20.0/24 is divided into three subnets of different sizes according to the requirements of the university. Lab A receives 62 usable host addresses, Lab B receives 30, and Lab C receives 14. The subnets are allocated without overlapping, and the remaining addresses from the original /24 block can be reserved for future expansion.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Introduction

Subnetting is the process of dividing a network into smaller logical networks called subnets. A subnet mask identifies the network portion and host portion of an IP address.

The given systems use the subnet mask 255.255.255.192, which corresponds to the CIDR prefix /26.

2. Given Data

Particulars	**System 1**	**System 2**
IP address	192.168.1.130	192.168.1.190
Subnet mask	255.255.255.192	255.255.255.192
CIDR notation	/26	/26

3. Subnetting Calculations

Step 1: Find the block size

The last octet of the subnet mask is 192.

{Block size}=256-192=64

Therefore, the subnet ranges increase in blocks of 64:

- 192.168.1.0 – 192.168.1.63
- 192.168.1.64 – 192.168.1.127
- 192.168.1.128 – 192.168.1.191
- 192.168.1.192 – 192.168.1.255

Both IP addresses, .130 and .190, fall within the third block, 192.168.1.128 – 192.168.1.191.

4. Analysis of System 1

Given IP address: 192.168.1.130

The IP address 130 falls within the block from 128 to 191.

- Network address: 192.168.1.128
- First valid host: 192.168.1.129
- Last valid host: 192.168.1.190
- Broadcast address: 192.168.1.191

Number of host bits:

32-26=6

Total IP addresses:

2^6=64

Usable host addresses:

2^6-2=64-2=62

Result: System 1's IP address 192.168.1.130 is a valid host address within its subnet.

5. Analysis of System 2

Given IP address: 192.168.1.190

The IP address 190 also falls within the block from 128 to 191.

- Network address: 192.168.1.128
- First valid host: 192.168.1.129
- Last valid host: 192.168.1.190
- Broadcast address: 192.168.1.191

Number of host bits:

32-26=6

Total IP addresses:

2^6=64

Usable host addresses:

2^6-2=62

Result: System 2's IP address 192.168.1.190 is also a valid host address within the same subnet. It is the last usable host address.

6. Summary Table

Particulars	**System 1**	**System 2**
IP address	192.168.1.130	192.168.1.190
Subnet mask	255.255.255.192	255.255.255.192
Network address	192.168.1.128	192.168.1.128
Subnet range (including network and broadcast)	192.168.1.128–191	192.168.1.128–191
Valid host range	192.168.1.129–190	192.168.1.129–190
Broadcast address	192.168.1.191	192.168.1.191
Usable host addresses	62	62
IP validity	Valid	Valid

7. Conclusion

Both systems belong to the same subnet, 192.168.1.128/26, because their IP addresses fall within the same subnet range. Both IP addresses are valid host addresses and can communicate directly at the IP subnet level, provided the systems are properly connected and configured. Each subnet supports 62 usable host IP addresses.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

1. Introduction

Subnetting is the process of dividing a large IP network into smaller subnetworks. It helps an organization allocate IP addresses efficiently to different departments and manage network traffic.

The organization has been allocated the IP block 172.16.0.0/16. The administrator assigns different subnet masks to four departments: HR, Sales, IT, and Admin.

2. Given Data

Department	CIDR Prefix	Subnet Mask
HR	/25	255.255.255.128
Sales	/26	255.255.255.192
IT	/27	255.255.255.224
Admin	/28	255.255.255.240

The original network /16 has the subnet mask 255.255.0.0.

3. Subnetting Formulas

1. Host bits = 32-prefix length
2. Total IP addresses = 2^host bits

3. Usable host addresses = 2^host bits-2
4. Block size = (256-last-octet subnet mask value), when calculating the block size in the fourth octet.

To avoid overlap, the departments are assigned consecutive subnets starting from 172.16.0.0.

4. Calculation for HR Department – /25

Step 1: Find the number of host bits

32-25=7 host bits

Step 2: Calculate the total and usable addresses

Total addresses=2^7=128

Usable hosts=128-2=126

Step 3: Determine the subnet range

The subnet mask is 255.255.255.128.

Block size=256-128=128

The first subnet starts at 172.16.0.0.

- Network address: 172.16.0.0
- First valid host: 172.16.0.1
- Last valid host: 172.16.0.126
- Broadcast address: 172.16.0.127

Therefore, HR receives 126 usable host addresses.

5. Calculation for Sales Department – /26

Step 1: Find the number of host bits

32-26=6 host bits

Step 2: Calculate the total and usable addresses

\[\text{Total addresses}=2^6=64 \]

\[\text{Usable hosts}=64-2=\boxed{62} \]

Step 3: Determine the subnet range

The subnet mask is 255.255.255.192.

\[\text{Block size}=256-192=64 \]

HR occupies addresses through 172.16.0.127. The next available /26 subnet begins at 172.16.0.128.

- Network address: 172.16.0.128
- First valid host: 172.16.0.129
- Last valid host: 172.16.0.190
- Broadcast address: 172.16.0.191

Therefore, Sales receives 62 usable host addresses.

6. Calculation for IT Department – /27

Step 1: Find the number of host bits

\[32-27=5\text{ host bits} \]

Step 2: Calculate the total and usable addresses

\[\text{Total addresses}=2^5=32 \]

\[\text{Usable hosts}=32-2=\boxed{30} \]

Step 3: Determine the subnet range

The subnet mask is 255.255.255.224.

\[\text{Block size}=256-224=32 \]

The next available /27 subnet begins at 172.16.0.192.

- Network address: 172.16.0.192
- First valid host: 172.16.0.193
- Last valid host: 172.16.0.222
- Broadcast address: 172.16.0.223

Therefore, IT receives 30 usable host addresses.

7. Calculation for Admin Department – /28

Step 1: Find the number of host bits

\[32-28=4\text{ host bits} \]

Step 2: Calculate the total and usable addresses

\[\text{Total addresses}=2^4=16 \]

\[\text{Usable hosts}=16-2=\boxed{14} \]

Step 3: Determine the subnet range

The subnet mask is 255.255.255.240.

\[\text{Block size}=256-240=16 \]

The next available /28 subnet begins at 172.16.0.224.

- Network address: 172.16.0.224
- First valid host: 172.16.0.225
- Last valid host: 172.16.0.238
- Broadcast address: 172.16.0.239

Therefore, Admin receives 14 usable host addresses.

8. Summary Table

Particulars	**HR**	**Sales**	**IT**	**Admin**
CIDR prefix	/25	/26	/27	/28
Subnet mask	255.255.255.128	255.255.255.192	255.255.255.224	255.255.255.240
Network address	172.16.0.0	172.16.0.128	172.16.0.192	172.16.0.224
First valid host	172.16.0.1	172.16.0.129	172.16.0.193	172.16.0.225
Last valid host	172.16.0.126	172.16.0.190	172.16.0.222	172.16.0.238
Broadcast address	172.16.0.127	172.16.0.191	172.16.0.223	172.16.0.239
Total IP addresses	128	64	32	16
Usable host addresses	126	62	30	14

9. Calculation of Remaining Unused IP Addresses

The original network block is 172.16.0.0/16.

Total addresses in the /16 block:

\[2^{32-16}=2^{16}=\boxed{65,536} \]

Total addresses allocated to the four departments:

\[128+64+32+16=240 \]

Therefore, the number of unallocated IP addresses is:

\[65,536-240=\boxed{65,296} \]

The allocated subnets occupy the address range 172.16.0.0 to 172.16.0.239.

Hence, the remaining unused IP address range is:

172.16.0.240 to 172.16.255.255

These addresses remain available for future departmental expansion or additional subnets.

10. Conclusion

The organization’s 172.16.0.0/16 network is divided into four non-overlapping subnets according to the assigned prefix lengths. HR receives 126 usable host addresses, Sales receives 62, IT receives 30, and Admin receives 14. In total, 240 IP addresses are allocated to the four departments, leaving 65,296 IP addresses unallocated for future use.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

1. Introduction

Subnetting is the process of dividing a large network into smaller networks called subnets. It helps to use IP addresses efficiently and improves network management.

The company has the network address 172.16.10.0/24 and uses a /27 subnet mask for its internal departments.

2. Given Data

Particulars	Details
Original network address	172.16.10.0/24
New subnet prefix	/27
Original prefix	/24
Subnet mask	255.255.255.224

3. Calculate the Total Number of Subnets

The number of bits borrowed from the host portion is:

$$ 27-24=3\text{ bits} $$

The formula for the number of subnets is:

$$ \text{Number of subnets}=2^{\text{borrowed bits}} $$

$$ =2^3=\boxed{8\text{ subnets}} $$

Therefore, 8 subnets are created.

4. Calculate the Number of Usable Hosts per Subnet

The number of host bits is:

\[32-27=5\text{ bits} \]

Total IP addresses per subnet:

\[2^5=32 \]

Usable host addresses:

\[2^5-2=32-2=\boxed{30} \]

Therefore, each subnet supports 30 usable host addresses.

5. Determine the Subnet of IP Address 172.16.10.78

The subnet mask is 255.255.255.224.

Calculate the block size:

\[\text{Block size}=256-224=32 \]

The subnet ranges increase in blocks of 32.

Subnet	Network address	Broadcast address
1	172.16.10.0	172.16.10.31
2	172.16.10.32	172.16.10.63
3	172.16.10.64	172.16.10.95
4	172.16.10.96	172.16.10.127
5	172.16.10.128	172.16.10.159
6	172.16.10.160	172.16.10.191
7	172.16.10.192	172.16.10.223
8	172.16.10.224	172.16.10.255

The IP address 172.16.10.78 lies between 172.16.10.64 and 172.16.10.95.

Therefore:

- Network address: 172.16.10.64
- First valid host: 172.16.10.65
- Last valid host: 172.16.10.94

- Broadcast address: 172.16.10.95

Hence, 172.16.10.78 belongs to the third subnet, 172.16.10.64/27.

6. Check Whether 172.16.10.95 Belongs to the Same Subnet

The IP address 172.16.10.95 is the broadcast address of the third subnet.

The valid host range of the third subnet is:

172.16.10.65 – 172.16.10.94

Since 172.16.10.95 is the broadcast address, it cannot be assigned to an individual host. It belongs to the same subnet range as 172.16.10.78, but it is not a valid host address.

7. Summary

Requirement	**Answer**
Total number of subnets	8
Usable hosts per subnet	30
Subnet containing 172.16.10.78	172.16.10.64/27
Network address	172.16.10.64
Valid host range	172.16.10.65 – 172.16.10.94
Broadcast address	172.16.10.95
Is 172.16.10.95 in the same subnet?	Yes
Can 172.16.10.95 be assigned to a host?	No, it is the broadcast address

8. Conclusion

The /27 subnet mask divides the original /24 network into 8 subnets, each supporting 30 usable host addresses. The IP address 172.16.10.78 belongs to the third subnet, with network address 172.16.10.64 and broadcast address 172.16.10.95. The address 172.16.10.95 belongs to this subnet but is reserved for broadcasting and cannot be assigned to an individual host.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

1. Introduction

IPv6 (Internet Protocol Version 6) is the latest version of the Internet Protocol. It uses 128-bit addresses, represented as eight groups of hexadecimal digits separated by colons.

The given IPv6 address is:

2001:0db8:0000:0000:0000:ff00:0042:8329

The prefix length is /64.

2. Compress the IPv6 Address

IPv6 addresses can be shortened using two rules:

1. Remove leading zeros: Zeros at the beginning of each hexadecimal group can be removed. For example, 0db8 becomes db8, and 0042 becomes 42.
2. Replace consecutive zero groups with ::: A single continuous sequence of one or more all-zero groups can be replaced by :: once in an address. Use the longest sequence when there is a choice.

Applying these rules:

Original address:

2001:0db8:0000:0000:0000:ff00:0042:8329

After removing leading zeros:

2001:db8:0:0:0:ff00:42:8329

The three consecutive zero groups can be replaced with ::.

Shortest compressed form:

\[\boxed{\texttt{2001:db8::ff00:42:8329}} \]

3. Expand the Compressed IPv6 Address

The compressed address is:

2001:db8::ff00:42:8329

To expand it:

- Replace db8 with 0db8.
- Replace 42 with 0042.
- The :: represents three missing zero groups, restoring the address to eight groups.

Full expanded form:

2001:0db8:0000:0000:0000:ff00:0042:8329

4. Determine the Network Prefix Using /64

The /64 prefix means that the first 64 bits represent the network portion, while the remaining 64 bits represent the interface identifier.

An IPv6 address contains eight groups of 16 bits each.

64 div 16=4 groups

Therefore, the first four groups form the network prefix.

Given address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Network portion: 2001:0db8:0000:0000

Network prefix in full notation:

2001:0db8:0000:0000::/64

Network prefix in compressed notation:

2001:db8::/64

The remaining four groups form the 64-bit interface identifier.

5. Calculate the Total Number of Possible Host Addresses

The total IPv6 address length is 128 bits.

Given prefix length = 64 bits.

Number of interface identifier bits:

128-64=64

The total number of possible addresses is:

2^64

=18,446,744,073,709,551,616

Therefore, the network supports:

2^64 possible addresses

IPv6 does not use broadcast addresses, unlike IPv4. Hence, the IPv4 rule of subtracting two addresses is not applied here.

6. Summary Table

Requirement	Answer
Original IPv6 address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed address	2001:db8::ff00:42:8329
Expanded address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix length	/64
Network prefix	2001:db8::/64
Interface identifier length	64 bits
Total possible addresses	2^64
Numerical total	18,446,744,073,709,551,616

7. Conclusion

The given IPv6 address is compressed to 2001:db8::ff00:42:8329 by removing leading zeros and replacing consecutive zero groups with ::. Its expanded form restores all eight groups. With a /64 prefix, the network prefix is 2001:db8::/64, leaving 64 bits for the interface identifier and providing 2^64 possible addresses within the network.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation